

DAY 1 · MODULE 2 · 45 MIN

Foundry Portal Tour

Projects, models, deployments, connections, Toolbox, Foundry IQ

What you'll leave with

- Navigate a Foundry project confidently
- Find and deploy a model
- Test a model in the playground
- Locate your project endpoint (you'll paste this into your `.env` in the lab)
- Recognize Toolbox and Foundry IQ (deep-dive comes Days 2–3)

Foundry resource architecture

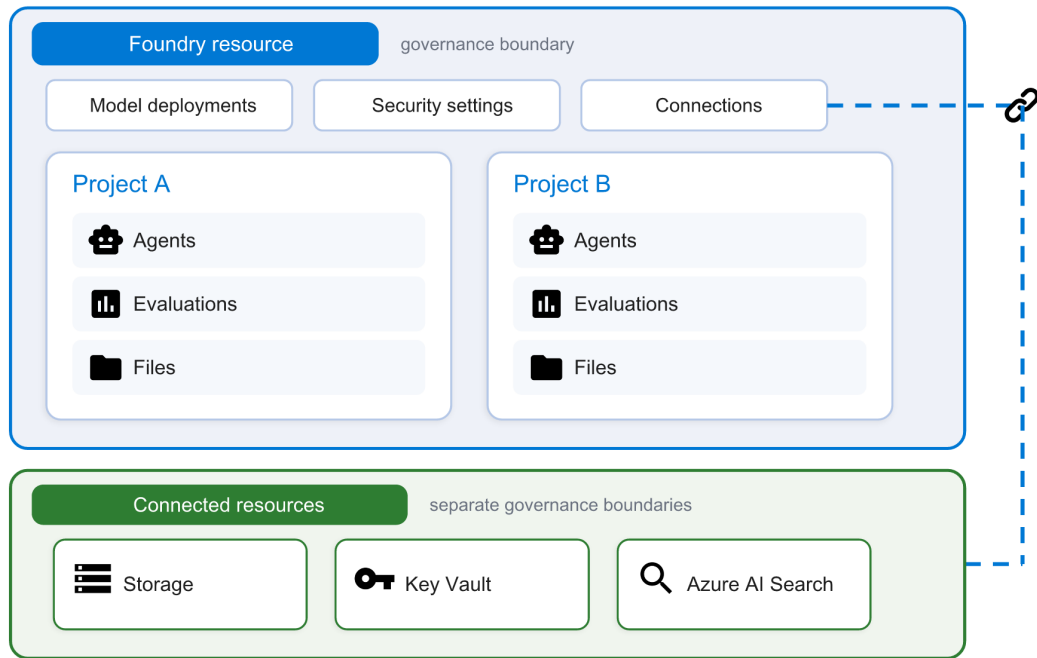


Diagram: Microsoft Learn · aka.ms/foundry/architecture

Four layers to know: Foundry resource → project → project assets → connected resources. Connected resources have their own networking and access policies.

D E M O

~5 min

Create a Foundry project, live

Portal path from + New project → App Insights checkbox → gpt-5.6-luna deployment → project endpoint visible. Then a 90-second look at the az CLI equivalent for the IaC-first path. Attendees who haven't done pre-work realize this is a 5-minute click-through, not a day of Azure ceremony.

Runbook: [demos/day1/module-2-demo-1-create-project.md](#)

What lives where · starter RBAC

Foundry resource level

- Model deployments
- Security & networking
- Connections to connected resources
- Resource-scoped RBAC roles (Foundry User, Foundry Owner)

Project level

- Agents (Prompt agents, Hosted agents)
- Files
- Evaluations
- Project-scoped RBAC assignments

Starter RBAC: assign every developer **Foundry User** at the Foundry resource scope. That covers Day 1 lab access. Fine-grained project-scoped roles come later.

Project endpoints

Every Foundry project has an endpoint like:

```
https://<foundry-  
resource>.services.ai.azure.com/api/projects/<your-project>
```

Set this as `FOUNDRY_PROJECT_ENDPOINT` in every lab's `.env`. MAF connects with this endpoint plus a credential — `AzureCliCredential` in dev, managed identity in production.

Model catalog

- Browse frontier models (OpenAI, Meta, Mistral, others by region)
- Filter by capability — chat, embeddings, vision, etc.
- See region availability, pricing tier, context window
- Not every model is available in every region — check region and quota first

Model deployments

- A deployment is a named, callable instance of a specific model
- Each deployment has:
 - Deployment name — what you pass to MAF as model=...
 - Model version
 - Capacity — tokens per minute
 - Region — inherited from the project
- Rule of thumb: one deployment per (model, role) pair. Don't share prod and dev.

Playgrounds

- Chat playground — test a deployed model interactively
- Agent playground — test a Prompt agent or Hosted agent you've created in Foundry
- Prompt flow / evaluation playgrounds — for testing eval configurations
- Use playgrounds to sanity-check before writing agent code
- If it doesn't work in the playground, it won't work in MAF either

Connections

- How a Foundry resource references other Azure services — AI Search, Storage, Key Vault, Fabric, and more
- Each connected resource is a separate Azure resource with its own networking and access policies
- Configured on the Foundry resource; projects inherit them
- Auth via managed identity or service principal — never long-lived keys in production
- We wire connections for Foundry IQ knowledge sources on Day 2

Toolbox and Foundry IQ

Introductions only — Days 2–3 go deep

Foundry Toolbox (intro)

- Curated catalog of ready-to-use tools an agent can attach to
- Includes: code interpreter, file search, Bing search, SharePoint, Fabric, Graph, custom skills
- Exposed to MAF over an MCP endpoint (same protocol as Day 3)
- Attach Toolbox tools to a Foundry-hosted agent from the portal, or from code

Foundry IQ (intro)

- The knowledge / grounding layer of Foundry
- Unified retrieval across enterprise sources — AI Search indexes, SharePoint, OneLake / Fabric, and more
- Alternative to hand-rolling a RAG pipeline per source
- Agents that use it get grounded answers with citations, without you writing retrieval code

Observability

- Every agent run in Foundry emits a trace — no code to enable, built-in for Prompt and Hosted agents
- A trace records: model calls, tool invocations, decisions, latency, tokens, and errors
- View traces in the Foundry portal (Tracing / Observability), or ship them to Application Insights
- OpenTelemetry semantics under the hood — same spans you'd expect from any OTel-instrumented service
- You'll open a real trace in Part B of today's lab; Day 5 goes deep on production observability

Two axes to remember

Axis	Feature	What it does
Actions — how the agent does things	Toolbox, MCP, function tools	Callable capabilities
Knowledge — how the agent knows things	Foundry IQ, RAG	Retrieval and grounding

Portal for learning · CLI for production

The portal is a great teacher and a great debugger. For creating resources and shipping changes, real teams use code.

Use the portal for

- Exploring what Foundry can do
- Sanity-checking a prompt in the playground
- Inspecting a trace when something goes wrong
- Monitoring dashboards and metrics

Use az / azd / SDK for

- Creating projects, deployments, agents
- Wiring connections to Storage / Key Vault / AI Search
- Deploying Hosted agents (zip + Foundry portal is a shortcut; azd / az is the norm)
- Anything that needs a repeatable, reviewable change

laC-first operating norm: resource creation and programmatic operations live in code. Every lab this week reflects that.

Common portal gotchas

- Quota errors on deployment — request a bump in the region before Day 1
- Missing role assignments — you need at least Foundry User (previously Azure AI User) at the Foundry resource scope; 401/403 usually means this
- Region mismatch — model deployment region must be reachable by IQ / AI Search
- Preview features move — Toolbox and parts of IQ are evolving; trust running code over screenshots

Live walkthrough (in the portal now)

- Sign into ai.azure.com and open a project
- Deploy a model
- Copy the project endpoint
- Open the chat playground; send a test prompt
- Show the Toolbox catalog and one entry
- Show the Foundry IQ landing page and one pre-provisioned knowledge source

Try it yourself: do steps 1–4 on your own sub before the module ends.

Takeaways

- Everything in Foundry lives inside a project.
- Copy your project endpoint and one model deployment name — you'll need both in every lab.
- Toolbox and Foundry IQ are today's introductions; Days 2–3 make them real.
- If portal pages don't work: quota → RBAC → region, in that order.

Next: Prompt engineering fundamentals — how to talk to models before we wrap them in agents.